PHC 6020, Spring 2026: Final Exam

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Problem 1.

a. This is a cross-sectional cohort study, which is an observational study. The exposure variable is smoking status, and the disease variable is liver cancer status.

b.

Relative risk is the risk in the exposed group over the risk in the unexposed group. We find the risk by looking at the proportion of smokers with liver cancer (30%), and the proportions of smokers without liver cancer (10%). This gives us:

$$\frac{Risk\ for\ exposed\ group}{Risk\ for\ unexposed\ group}=\frac{\frac{90}{300}}{\frac{70}{700}}=\frac{0.3}{0.1}=3$$

Smokers have 3 times the chance of developing liver cancer as opposed to non-smokers.

We now calculate the odds ratio, which is the odds of liver cancer in smokers vs. the odds of liver cancer in non-smokers:

$$Odds\ Ratio=\frac{\frac{90}{210}}{\frac{70}{630}}=\frac{0.428}{0.111}=3.86$$

The odds of having liver cancer are 3.86 higher for smokers than they are for non smokers.

c.

	Having liver cancer	Not having liver cancer	Total
Smoker	90	210	300
Non-smoker	70	630	700
Total	160	840	1000

First we find the standard error of the odds ratio:

$$SE=\sqrt{\frac{1}{90}+\frac{1}{210}+\frac{1}{70}+\frac{1}{630}}=0.178$$

Now for the 95% CI for the log odds ratio

$$\log(3.86)\pm 1.96*0.178$$

$(1.00, 1.70)$

It appears that the relationship between smoking and liver cancer is significant in this population, as our 95% confidence interval does not pass through zero.

Problem 2.

a.

This is another cross-sectional cohort study. The exposure variable is smoking status, and the disease variable is lung cancer status.

b.

	Having lung cancer	Not having lung cancer	Total
Smoker	80	120	200
Non-smoker	40	760	800
Total	120	880	1000

We calculate the odds ratio:

$$\text{OR} = \frac{\frac{80}{120}}{\frac{40}{760}} = \frac{0.667}{0.053} = 12.6$$

The odds of smokers in this sample having lung cancer is 12.6 times higher for the smoking group than the non-smoking group.

We find the standard error next:

$$SE = \sqrt{\frac{1}{80} + \frac{1}{120} + \frac{1}{40} + \frac{1}{760}} = 0.217$$

So, the 95% CI for the log odds ratio is:

$$\log(12.6) \pm 1.96 * 0.217 = 2.53 \pm 0.425 = (2.10, 2.96)$$

The relationship between smoking and lung cancer appears to be significant as our 95% confidence interval does not pass through zero.

Problem 3.

To perform stratified randomization, you form strata using various factors (age, gender, race, etc.) then use permuted block randomization in each strata as patients enroll. For example, if you stratify by gender, you randomize within the Male strata and the Female strata. The advantage of stratified randomization is making the treatment groups similar, and allowing better estimates of treatment difference due to the strata. The disadvantage is that if you have too many prognostic factors, you could end up with too few patients in some strata, which would be a weak design.

Problem 4.

a.

We will determine the sample size as follows:

$z_{\frac{\alpha}{2}}=1.96$ for two-sided

$z_{\beta}=0.84$ for 80% power

$\sigma_Y=5$

$\Delta=2$

$$n=\left(\frac{(Z_{\alpha/2}+Z_{\beta})^2*\sigma_Y^2*4}{\Delta^2}\right)$$

$$n=\frac{(1.96+0.84)^2*5^2*4}{2^2}=\frac{7.84*25*4}{4}=196$$

b.

We will determine the power if 100 patients are used as follows. First, we rearrange the formula solving for power.

$n=\left(\frac{(Z_{\alpha/2}+Z_{\beta})^2*\sigma_Y^2*4}{\Delta^2}\right)$ and $P(T\geq z_{\alpha}|\Delta=\Delta_A)=1-\beta$ for determining power.

$$\frac{\Delta_A}{\sigma_Y}*\sqrt{\frac{2}{n}}=Z_{\alpha/2}+Z_{\beta}$$

We now solve for Z_{β}

$$\frac{2}{5}*\sqrt{\frac{2}{100}}=1.96+Z_{\beta}$$

$$Z_{\beta}=0.05656-1.96=-1.903$$

Using R, we determine β with the critical value -1.903:

```
> pnorm(-1.903)
[1] 0.02852027
```

So, our power is 1 – 0.029 = 0.971.

With 100 patients in each group, the power is 97.1%

Problem 5.

a.

First, we construct a table of the expected counts:

$$E_{yes}=\frac{500*500}{1500}=166.67$$

$$E_{no}=\frac{1000*500}{1500}=333.33$$

Response	Treatment 1	Treatment 2	Treatment 3	Total
yes	166.67	166.67	166.67	500
no	333.33	333.33	333.33	1000
Total	500	500	500	1500

The chi-square test statistic formula is:

$$X^2=\frac{\sum_{cells}(O_j-E_j)^2}{E_j}$$

$$X^2=\frac{(150-166.67)^2}{166.67}+\frac{(350-333.33)^2}{333.33}+\frac{(125-166.67)^2}{166.67}+\frac{(375-333.33)^2}{333.33}+\frac{(225-166.67)^2}{166.67}+\frac{(275-333.33)^2}{333.33}$$

$$X^2=1.67+0.83+10.42+5.21+20.41+10.21=48.75$$

Our degrees of freedom is 2, noted by (2rows – 1)(3columns – 1) = 1 * 2

So, we obtain our critical value from a chi-square distribution table, or we can use R:

```
> 1 - pchisq(48.75, 2)
[1] 2.594613e-11
```

This is less than our level of 0.05 per the instructions, so we determine that H_A is not all the same as H_0.

b.

We apply Bonferroni correction to accomplish this problem. We calculate our response rates for the 3 treatments:

$$p_1 = \frac{150}{500} = 0.3$$

$$p_2 = \frac{125}{500} = 0.25$$

$$p_3 = \frac{225}{500} = 0.45$$

We now use Bonferroni correction for 3 pairwise comparisons, so our alpha is:

$\alpha = \frac{0.05}{2} = 0.025$ and corresponding critical value is $z_{\alpha/2} = 2.24$

```
> qnorm(1 - 0.025/2)
[1] 2.241403
```

So, we are interested if $T_{1j} \geq 2.24$.

$$T_{1j} = 2\frac{\sqrt{n_1 * n_j}}{n_1 + n_j}\left(\sin^{-1}\sqrt{p_j} - \sin^{-1}\sqrt{p_1}\right)$$

Comparing treatment 1 and 2:

$$T_{12} = 2\sqrt{\frac{500*500}{500+500}}\left(\sin^{-1}\sqrt{0.25} - \sin^{-1}\sqrt{0.30}\right) = 2*15.81(0.580 - 0.524) = 1.771$$

1.771 is less than 2.24, so the treatments are not significantly different. We now compare treatment 1 and 3:

$$T_{12} = 2\sqrt{\frac{500*500}{500+500}}\left(\sin^{-1}\sqrt{0.45} - \sin^{-1}\sqrt{0.30}\right) = 2*15.81(0.735 - 0.580) = 1.771 = 4.901.$$

4.901 is greater than 2.24, so treatments 1 and 3 are significantly different from one another, with treatment 3 being a more effective choice.

Problem 6.

a.

The response rate of the brand drug is 30%, and the FDA requires the response rate to be not 5% worse, so our δ is 0.05. Our Δ is the difference in response of the generic drug and the brand drug, or, $\mu_G - \mu_B$

For non-inferiority hypothesis test, we have:

$H_0 : \Delta \leq -\delta$

$H_A : \Delta > -\delta$ for delta > 0.

Therefore, our hypothesis test is:

$H_0 : \mu_G - \mu_B \leq -0.05$

$H_1 : \mu_G - \mu_B > -0.05$

b.

Per the set up of the problem, we have:

$\delta = 0.05$

Brand response rate (μ_B) = 0.30

$\mu_G = 0.3$

95% power Z-stat: $z_{0.05} = 1.64$

For two sample t-test with equal allocation, we have:

$$n = \frac{(z_\alpha + z_\beta)^2 \sigma_Y^2 * 4}{\Delta_A^2}$$

We have:

$\Delta_A = \mu_G - \mu_B + \delta = 0.05$ and $\sigma_Y^2 = \mu_G(1-\mu_G) + \mu_B(1-\mu_B) = 0.3(0.7) + 0.3(0.7) = 0.42$

So, we have:

$$n = \frac{(1.64 + 1.64)^2 * 0.42 * 4}{0.05^2} = \frac{18.07}{.0025} = 7228$$

Problem 7.

a.

DSMB stands for Data Safety Monitoring Board.

b.

The role of DSMB in clinical trials is to ensure the safety and well-being of people participating in the trial. They should not have a conflict of interest with the studies they monitor. They review protocols, study progress, safety, data quality, and efficacy and benefit. They are a very important aspect of a clinical trial.

c.

The Data Safety Monitoring Board is needed in most Phase III clinical trials, multicenter and randomized Phase II studies and Phase II studies that have a high risk factor. DSMB is also critical when the participants are extra vulnerable.

Problem 8.

a.

We have hazard function $\lambda(t)=\frac{\frac{-dS(t)}{dt}}{S(t)}$

We are given the hazard function:

$\lambda(t)=\gamma t$

So, we have:

$\frac{-dS(t)}{dt}=\gamma t S(t)$ which rearranged equals $\frac{dS(t)}{S(t)}=-\gamma t\, dt$

We can integrate both sides:

$\int \frac{dS(t)}{S(t)}=\int -\gamma t\, dt$ which gives us $\ln|S(t)|=\frac{-\gamma t^2}{2}+C$, with C being the integration constant. We want to solve for S(t) now, so we exponentiate both sides:

$S(t)=e^{\frac{-\gamma t^2}{2}+C}$. We set S(0) = 1 to determine integration constant C:

$S(0)=e^{\frac{-\gamma 0^2}{2}+C}=e^C=1$, so plugging e^C back in we have:

$S(t)=e^{-\frac{\gamma t^2}{2}}$ as the expression of the survival function $S(t)$

b.

We derive expressions of the median survival time as follows:

S(t) = 0.5.

So, with our survival function S(t):

$S(t)=\exp(-\int_0^t \lambda(u)\,du)$ and the given hazard, we have:

$S(t)=\exp(-\int_0^t \gamma u\,du)$ and can solve the integral: $\int_0^t \gamma u\,du=\frac{\gamma * t^2}{2}$ and get:

$S(t)=\exp(\frac{-\gamma t^2}{2})$. We set t = 0.5 for median survival time:

$0.5=\exp(\frac{-\gamma t_{median}^2}{2})$ and take natural log of both sides:

$\ln(0.5)=\frac{-\gamma t_{median}^2}{2}$ To make solving for t easier, ln(0.5) = -ln(2), so we can substitute. And then we solve for t:

$t_{median}=\sqrt{\frac{2\ln(2)}{\gamma}}$

Problem 9.

a.

9

part (a):

Interval [0,10): , Survival Prob.: $P(10) = 1 - \frac{1}{10} = .9$

- Patient 8 , Survival function at $t=10$: $S(10) = 1 \times .9 = .9$
- 1 event
- 10 patients at risk

$S(10) = .9$

Interval [10, 20): , $P(20) = 1 - \frac{1}{9} = .89$

- Patient 12, 16+, 19+, $t=20$: $S(20) = S(10) \times .89 = .9 \times .89 = 0.801$
- 1 event
- 9 patients at risk after censor

$S(20) = .801$

Interval [20, 30): , $P(30) = 1 - \frac{2}{7} = .714$

- Patient 23, 26, 26+, 29+, $t=30$: $S(30) = S(20) \times .714 = .801 \times .714 = 0.572$
- 2 events
- 7 patients at risk

$S(30) = .572$

Interval [30, 40), $P(40) = 1 - \frac{1}{6} = .83$

- Patient 33+, 36, $t=40$, $S(40) = S(30) \times .83 = .572 \times .83 = .689$
- 1 event
- 6 patients at risk

$S(40) = .689$

b.

Part b:

We find standard error for S(10):

$$se\{\hat{S}(t)\} = \hat{S}(t)\left\{\sum_{j=1}^{t} \frac{d_j}{(n_j - \frac{w_j}{2})(n_j - d_j - \frac{w_j}{2})}\right\}^{1/2}$$

$$se(S(10)) = 0.9\sqrt{\frac{1}{(10-0)(10-1-0)}}$$

$$= .9\sqrt{\frac{1}{90}} = .9 \times \frac{1}{\sqrt{90}} \doteq .9 \times .105 = .0945$$

95% confidence interval:

$$\hat{S}(t) \pm Z_{\alpha/2}(se(\hat{S}(t)))$$

$$.9 \pm 1.96 \times .0945 = .9 \pm .185 = \boxed{(.715,\ 1.085)}$$

$$Se(S(20)) = .801\sqrt{\frac{1}{(9-\frac{1}{2})(9-2-\frac{1}{2})} + \frac{1}{(10)(10-1)}} = .801 \times \frac{1}{\sqrt{55.25}} + \frac{1}{\sqrt{90}}$$

$$= .801 \times \frac{1}{7.43} \times \frac{1}{9.49} = .011$$

95% confidence interval:

$$.801 \pm 1.96 \times .011 = .801 \pm .022 = \boxed{(.779,\ .823)}$$

Problem 10.

Generally speaking, intent-to-treat analysis (ITT) ignores the fact that some patients in the trial do not wind up receiving the treatment for one reason or another and analyzes the data based on the originally assigned groups. This means that even if the patient refuses the assigned treatment, does not follow instructions for dosage, changes the treatment, or falls under the category of noncompliance for another capacity, that patient is still included in analysis. The advantage of this is that the strengths of randomization are preserved even in the event of noncompliance. Also, it agrees with per-protocol analysis. It can also capture the effect of the difficulty of using the treatments. On the other hand, a disadvantage of ITT is that it might be better suited at comparing policies more than the effectiveness of a treatment, which in some cases might not be the right approach. There is also the higher chance of dealing with missing data compared to other analysis types.